

BIJAY THAKUR

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LinkedIn

GitHub

Portfolio

SUMMARY

AI/ML builder focused on RAG, LLM applications, agentic workflows, and backend systems. Built evaluated GenAI projects with OpenAI, Gemini, LangGraph, LlamaIndex, ChromaDB, Hugging Face, FastAPI, and Next.js. Ex AI Agent Builder Intern at NeuralSeek.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Java, C++, SQL
- **AI & ML:** RAG, Agentic AI, Multi-Agent Systems, Fine-Tuning, NLP, Transformers, Hugging Face, Prompt Engineering, PyTorch, Model Evaluation, Embeddings, Deep Learning, Machine Learning
- **Backend/DevOps:** FastAPI, Flask, REST APIs, Docker, GitHub Actions, ChromaDB, Vector Databases

PROJECTS

SEC 10-K RAG - Evaluated Financial QA System | GitHub April 2026 - May 2026

- Built a citation-grounded **SEC 10-K RAG** pipeline with structured HTML/XBRL parsing, **OpenAI embeddings**, **ChromaDB**, **BM25**, **hybrid RRF retrieval**, and cross-encoder reranking over Apple's 2025 annual report.
- Implemented a parallel **LlamaIndex** version to compare framework vs. from-scratch **RAG** design, reducing core pipeline code by **22%** while achieving **0.74 Recall@1**, **0.8032 MRR**, and **0.9901 RAGAS** faithfulness on a 50-question gold eval set.

LaunchLab AI - Agentic Multi-LLM Product Studio | GitHub April 2026 - May 2026

- Built a **multi-LLM** product studio using **Next.js**, **Flask**, **Gemini**, **OpenAI**, and **LangGraph** to convert rough product ideas into research, product blueprints, **Claude Code** build prompts, campaign copy, visual prompts, and evaluation reports.
- Implemented shared-state agent orchestration, provider fallbacks, structured error handling, and workflow visualization to compare manual pipelines against scalable **LangGraph** execution.

MarianMT Domain Adaptation - English-to-French Neural Machine Translation May 2026

- Fine-tuned **Helsinki-NLP/opus-mt-en-fr** with **Hugging Face Transformers**, **PyTorch**, and **CUDA GPU** on **Google Colab**, scaling experiments across 1K, 10K, and 189K KDE4 examples for **3 epochs** and improving English to French translation **BLEU from 46.07 to 58.37**.
- Built an end-to-end **Hugging Face Seq2SeqTrainer** workflow with **tokenization**, **DataCollatorForSeq2Seq**, **mixed-precision GPU** training, checkpoint saving, and per-epoch **BLEU/eval-loss** tracking, validating that larger in-domain **KDE4** training sets consistently improved translation quality across all experiments.

EXPERIENCE

AI Agent Builder Intern, NeuralSeek April 2026 - May 2026

Remote

- Built and refined multi-step AI agent workflows for reasoning, automation, and response validation, improving assistant behavior through prompt iteration, workflow testing, and failure analysis.
- Evaluated agent outputs across edge cases, identified reasoning and reliability gaps, and documented failure patterns to support more consistent user-facing AI workflows.

Academic Tutor, LaGuardia Community College Feb 2024 - Jun 2024

Long Island City, NY

- Tutored students 1-on-1 in C++, debugging, and algorithmic problem-solving while creating structured notes and practice modules to strengthen core CS fundamentals.

Web Developer Intern, WebX Nepal Mar 2022 - Jul 2022

Kathmandu, Nepal

- Developed responsive frontend pages and UI components using HTML, CSS, and JavaScript, collaborating with developers to translate client requirements into functional layouts and improve cross-device UX.

EDUCATION

Queens College, The City University of New York

Expected: Dec 2026

Bachelor of Science in Computer Science, GPA: 3.64